MA 102 Mathematics II

Ordinary Differential Equations

Lecture 20

(April 23, 2024)

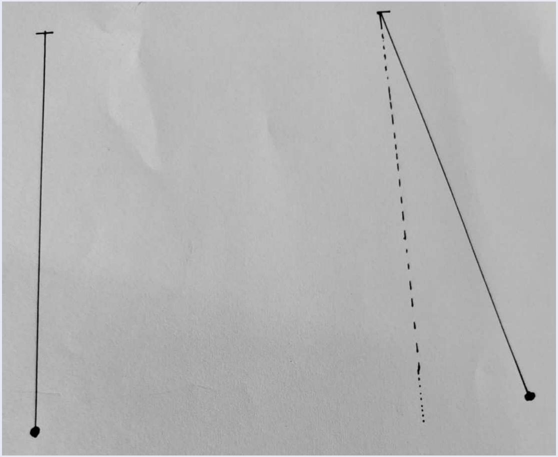

Figure 0

The above figure describes the stable and unstable nature of the steady state of the motion of a pendulum bob. (Discussed in the last lecture)

Recall

Consider the following autonomous system:

$$\left.\begin{array}{l} \frac{dx}{dt} = a_1 x + b_1 y, \\ \frac{dy}{dt} = a_2 x + b_2 y, \end{array}\right\} \tag{1}$$

which has $(0,0)$ as an obvious critical point.

We assume throughout that

$$\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} \neq 0 \tag{2}$$

so that $(0,0)$ is the only critical point.

The auxiliary equation is

$$m^2 - (a_1 + b_2)m + (a_1 b_2 - a_2 b_1) = 0. \tag{3}$$

Also recall

Theorem (Case A): If the roots m_1 and m_2 of (3) are real, distinct, and of the same sign, then the critical point $(0, 0)$ of the system (1) is a node.

Theorem (Case B): If the roots m_1 and m_2 of (3) are real, distinct, and of opposite signs, then the critical point $(0, 0)$ of the system (1) is a saddle point.

Theorem (Case C): If the roots m_1 and m_2 of (3) are complex conjugates but not purely imaginary, then the critical point $(0, 0)$ of the system (1) is a spiral.

Theorem (Case D)

If the roots m_1 and m_2 of (3) are real and equal, then the critical point $(0,0)$ of the system (1) is a node.

Proof

Assume that $m_1 = m_2 = m < 0$ for the auxiliary equation (3).

We have the following cases to discuss: (i) $a_1 = b_2 \neq 0$ and $a_2 = b_1 = 0$
(ii) all other possibilities leading to a double root of equation (3).

Case (i)

If a denotes the common value of a_1 and b_2, then (3) becomes $m^2 - 2am + a^2 = 0$ and hence $m = a$. The system (1) becomes

$$\left.\begin{aligned} \frac{dx}{dt} &= ax, \\ \frac{dy}{dt} &= ay. \end{aligned}\right\} \tag{4}$$

In this case, the general solution of (4) is given by

$$\left.\begin{aligned} x &= c_1 e^{mt}, \\ y &= c_2 e^{mt}, \end{aligned}\right\} \tag{5}$$

where c_1 and c_2 are arbitrary constants.

The paths defined by (5) are half-lines of all possible slopes, i.e., the y-axis for $c_1 = 0$, the x-axis for $c_2 = 0$, the lines $y = (c_2/c_1)x$ for $c_1 \neq 0, c_2 \neq 0$. Refer to Figure 1.

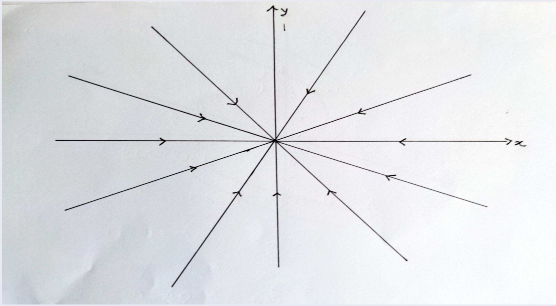

Figure 1

Since we have taken $m < 0$, it can be observed that each path approaches and enters $(0,0)$ as $t \to \infty$. Therefore, the critical point is a node. It is also asymptotically stable.

If we consider $m > 0$, then the situation will be the same except that the paths enter $(0,0)$ as $t \to -\infty$. The arrows in the figure will be reversed.

In this case, the node will be unstable.

It implies that the node can be stable as well as unstable which depends on the sign of the root of the auxiliary equation.

Case (ii)

In this case, we know that the general solution for $m_1 = m_2$ can be written in the form

$$\left.\begin{aligned} x &= c_1 A e^{mt} + c_2(A_1 + A_2 t)e^{mt}, \\ y &= c_1 B e^{mt} + c_2(B_1 + B_2 t)e^{mt}, \end{aligned}\right\} \tag{6}$$

where A's and B's are definite constants and c are arbitrary constants.

When $c_2 = 0$, we obtain

$$\left.\begin{aligned} x &= c_1 A e^{mt}, \\ y &= c_1 B e^{mt}. \end{aligned}\right\} \tag{7}$$

Solutions (7) represent two half-line paths lying on the line $Ay = Bx$ with slope B/A and since $m < 0$, both paths approach $(0, 0)$ as $t \to \infty$. Refer to Figure 2.
Further, since $y/x = B/A$, both paths enter $(0, 0)$ with slope B/A.

If $c_2 \neq 0$, the solutions (6) represent curved paths, and since $m < 0$, it is clear from (6) that these paths approach $(0, 0)$ as $t \to \infty$.

Further,

$$\frac{y}{x} = \frac{c_1 B e^{mt} + c_2(B_1 + B_2 t)e^{mt}}{c_1 A e^{mt} + c_2(A_1 + A_2 t)e^{mt}} = \frac{c_1 B/c_2 + B_1 + B_2 t}{c_1 A/c_2 + A_1 + A_2 t}$$

and hence it follows that $y/x \to B_2/A_2$ as $t \to \infty$ and all these curved paths enter $(0,0)$ with slope B_2/A_2.

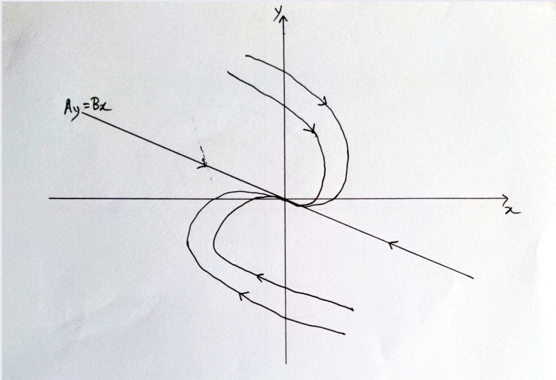

Figure 2

Figure 2 gives a clear picture of the arrangement of these paths. It is obvious that $(0, 0)$ is a node which is asymptotically stable.

Now, if $m > 0$, the situation remains unchanged except that the directions of the paths are reversed and the critical point is unstable.

Theorem (Case E)

If the roots m_1 and m_2 of (3) are purely imaginary, then the critical point $(0,0)$ of (1) is a center.

Proof

It has resemblance with Case C in which we considered the complex roots. Now the roots are of the form $a \pm ib$ with $a = 0$ and $b \neq 0$. It is clear that now the general solution will be given by

$$\left.\begin{aligned} x &= [c_1(A_1 \cos bt - A_2 \sin bt) + c_2(A_1 \sin bt + A_2 \cos bt)], \\ y &= [c_1(B_1 \cos bt - B_2 \sin bt) + c_2(B_1 \sin bt + B_2 \cos bt)]. \end{aligned}\right\} \tag{8}$$

It is clear that $x(t)$ and $y(t)$ are periodic and each path is a closed curve surrounding the origin. As seen from Figure 3, these curves represent ellipses. Therefore, the critical point $(0,0)$ is a center which is stable but not asymptotically stable.

That $x(t)$ and $y(t)$ represent ellipses can be achieved in two ways:
(i) by squaring x and y in (8) and adding them, or
(ii) by solving the differential equation

$$\frac{dy}{dx} = \frac{a_2 x + b_2 y}{a_1 x + b_1 y}.$$

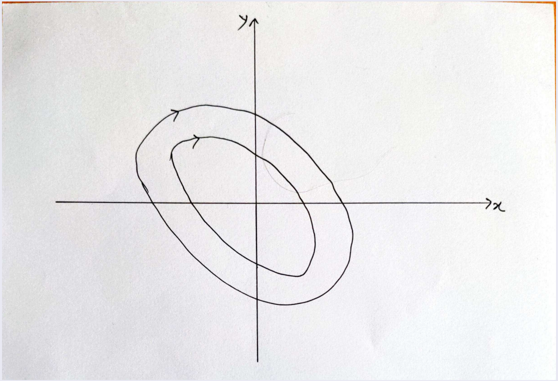

Figure 3

Theorem (Stability)

The critical point $(0,0)$ of the linear system (1) is stable if and only if both roots of the auxiliary equation (3) have non-positive real parts, and it is asymptotically stable if and only if both roots have negative real parts.

Observation

Let us write equation (3) in the form

$$(m - m_1)(m - m_2) = m^2 + pm + q = 0, \tag{9}$$

so that $p = -(m_1 + m_2)$ and $q = m_1 m_2$.

If we interpret our previous results in the *pq*-plane, then we arrive at a striking diagram that displays at a glance the nature and stability properties of the critical point $(0,0)$. (Figure 4)

The first observation is that the p-axis (i.e., $q = 0$) is excluded, since by condition (2) we know that $m_1 m_2 \neq 0$. We have

$$m_1, m_2 = \frac{-p \pm \sqrt{p^2 - 4q}}{2}.$$

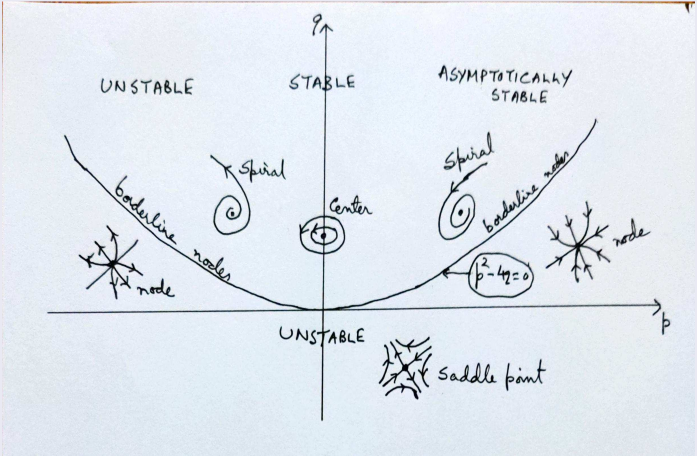
q
UNSTABLE
STABLE
ASYMPTOTICALLY STABLE
Spiral
Spiral
borderline nodes
borderline nodes
Center
node
node
$p^2-4q=0$
p
UNSTABLE
Saddle point

Figure 4

Thus above the parabola $p^2 - 4q = 0$, we have $p^2 - 4q < 0$, so m_1 and m_2 are conjugate complex numbers that are purely imaginary if and only if $p = 0$: these are cases C and E.

Below the p-axis, we have $q < 0$, which means that m_1 and m_2 are real, distinct and have opposite signs: this is case B.

Finally, the zone between these regions (including the parabola, excluding the p-axis) is characterized by the relations $p^2 - 4q \geq 0$ and $q > 0$, so m_1 and m_2 are real and of the same sign: these are cases A and D.

In addition, it is clear that there is precisely one region of asymptotic stability: the first quadrant. This can be stated as follows:

Theorem

The critical point $(0, 0)$ of the linear system (1) is asymptotically stable if and only if the coefficients $p = -(a_1 + b_2)$ and $q = a_1 b_2 - a_2 b_1$ of the auxiliary equation (3) are both positive.

Consider the system

$$\left.\begin{aligned} \frac{dx}{dt} &= F(x,y) \\ \frac{dy}{dt} &= G(x,y) \end{aligned}\right\}, \quad t \in (-\infty, \infty). \tag{10}$$

The main concern is the behaviour of the solution of (10) in the neighbourhood of the critical point, in particular $(0,0)$.

Many nonlinear systems can be approximated by a system like

$$\frac{d^2x}{dt^2} = f\left(x, \frac{dx}{dt}\right). \tag{11}$$

$(x(t), y(t))$ denotes the solution in (t, x, y)-space (known as motion space).

While studying the nature of the solution in the neighbourhood of $(0,0)$, we consider its behaviour only in the xy-plane. As t varies, $x = x(t),\ y = y(t)$ describe parametrically a curve in the xy-plane, known as path/trajectory.

Parametric form of a parabola:

$$C : (t^2/4a, t).$$

There is a difference between the solution of (11) and the trajectory of (11). Take $dx/dt = 1, dy/dt = 2$. This implies

$$x(t) = t + c_1,\ y(t) = 2t + c_2.$$

In the phase plane, $dy/dx = 2$ implies $y = 2x + c$ which is a straight line with slope 2. Clearly this solution does not have a critical point and several solutions may present the same trajectory.

If a neighbourhood N exists of the origin such that every solution starting in N approaches the origin as $t \to \infty$, then the origin is called an attractor.

If the trajectory starts in N and tends to the origin as $t \to -\infty$, then the origin is called a repeller.